CUSTOMER CHURN & RETENTION ANALYSIS

Internship Project | Subscription Customer Analytics

Executive Summary

This report analyzes 7,043 telecom customers to understand churn patterns and identify practical retention opportunities. 1,869 customers have churned (26.5%), while 5,174 remain retained (73.5%). Average tenure is 32.4 months. The strongest observed churn association is contract type: month-to-month customers churn at 42.7%, compared with 2.8% for two-year contracts.

Metric	Result
Customers	7,043
Churned	1,869 (26.5%)
Retained	5,174 (73.5%)
Average tenure	32.4 months
Average monthly charges	$64.76
Average total charges	$2279.73

Key Findings

1. Contract type: month-to-month churn is 42.7%; two-year contract churn is 2.8%. This is a strong descriptive association, though contract selection may reflect other customer differences.

2. Payment method: electronic-check customers churn at 45.3%; investigate payment friction, billing experience and segment mix.

3. Support: customers without tech support churn at 41.6%. This suggests support access or customer needs may be related to retention.

4. Security: customers without online security churn at 41.8%; test education and bundling before assuming this feature causes retention.

5. Lifetime pattern: churned customers have average tenure of 18.0 months, compared with 37.6 months among retained customers.

Recommendations

1. Reduce month-to-month churn. Offer transparent annual-plan benefits, with opt-in discounts and reminders before renewal.

2. Strengthen early lifecycle. Build a 30/60/90-day onboarding journey, proactive check-ins and early usage nudges.

3. Improve support coverage. Offer a support trial, proactive troubleshooting and clearer self-service help.

4. Review payment friction. Test opt-in autopay incentives and monitor failed/late payments; validate causality before changing policy.

5. Improve security value proposition. Test clearer security-feature education or bundles; compare like-for-like segments before concluding impact.

6. Build proper cohort tracking. Capture signup date, cancellation date, plan-change dates and monthly active status to create monthly cohort retention and survival curves.

Data Limitations

The dataset contains 7,043 customer-level records and includes tenure, contract and churn status, but no signup date, cancellation date, or monthly activity history. Therefore, true signup-month cohort retention, churn timing by calendar month, and time-to-event survival curves cannot be calculated reliably from this file alone. Tenure bands are used as a lifecycle proxy. 11 blank TotalCharges values were converted to zero after numeric conversion; this should be validated against source documentation.

Methodology

Churn rate = churned customers ÷ total customers. Retention rate = 1 − churn rate. Segment churn rates are calculated within each segment. Findings describe associations and should be validated with controlled experiments or further modeling before causal claims are made.